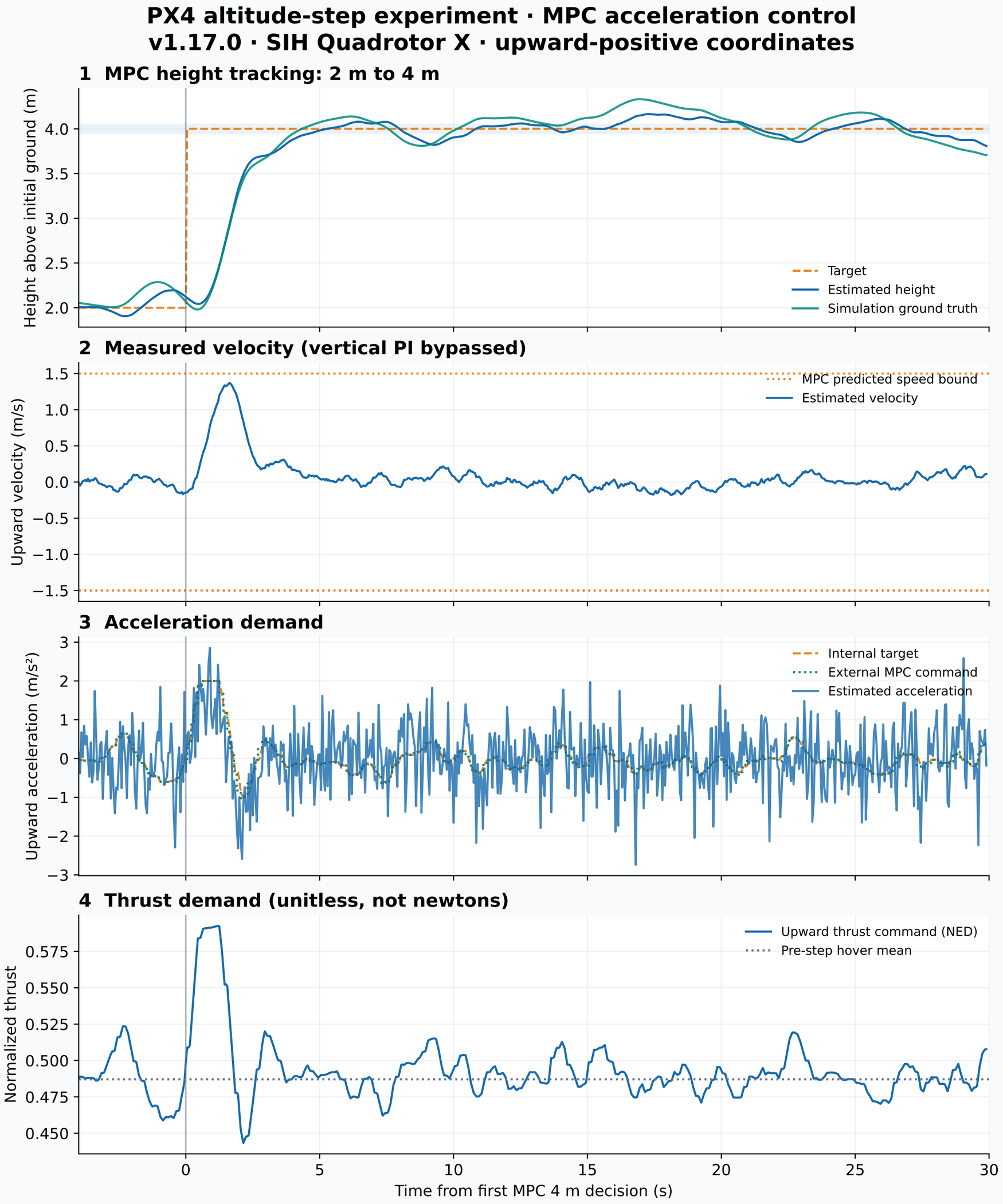

PX4 altitude-step experiment · MPC acceleration control
v1.17.0 · SIH Quadrotor X · upward-positive coordinates
1 MPC height tracking: 2 m to 4 m
Height above initial ground (m)
Target
Estimated height
Simulation ground truth
2 Measured velocity (vertical PI bypassed)
Upward velocity (m/s)
MPC predicted speed bound
Estimated velocity
3 Acceleration demand
Upward acceleration (m/s²)
Internal target
External MPC command
Estimated acceleration
4 Thrust demand (unitless, not newtons)
Normalized thrust
Upward thrust command (NED)
Pre-step hover mean
Time from first MPC 4 m decision (s)